THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): Red blood cells swell and burst in distilled water. Reason (R): Water enters the cells by osmosis because they lack a cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Grasses grow again after being cut. Reason (R): Intercalary meristem is present at the base of their leaves. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Whale is placed in the class Mammalia. Reason (R): It has mammary glands and breathes with lungs. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Plants raised by vegetative propagation are genetically identical to the parent. Reason (R): They are produced by mitosis from a single parent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): The polar regions are colder than the equator. Reason (R): Sunlight falls at a slant at the poles and is spread over a larger area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A cell is placed in a solution and its mass increases by 20 per cent. The solution is: 1

(a) saturated salt **(b)** isotonic
(c) hypertonic **(d)** hypotonic to the cell

7. A ring of bark is removed from a branch. After some days, the part above the ring swells due to: 1

(a) accumulation of water **(b)** minerals
(c) gas **(d)** accumulation of food

8. A new organism is found to be multicellular, with cell walls of chitin, non-motile, and with thread-like hyphae. Its kingdom is: 1

(a) Plantae **(b)** Animalia **(c)** Fungi **(d)** Protista

9. A woman's menstrual cycle is 28 days and menstruation begins on 1 March. The likely day of ovulation is: 1

(a) 28 March **(b)** 7 March **(c)** 1 March **(d)** 14 March

10. A surface of albedo 0.5 receives 800 W/m squared. The absorbed radiation per square metre in 10 s is: 1

(a) 800 J **(b)** 4000 J **(c)** 8000 J **(d)** 400 J

11. Which of the following could not be a function of the Golgi apparatus? 1

(a) formation of lysosomes **(b)** secretion
(c) synthesis of ATP **(d)** packaging proteins

12. The tissue that produces the enzymes and hormones in glands is: 1

(a) bone **(b)** muscle
(c) squamous epithelium **(d)** glandular epithelium

13. **Assertion (A):** Mitochondria are called the powerhouses of the cell. 1
Reason (R): They release energy from food through cellular respiration and store it in the form of ATP.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Meristematic tissue is responsible for the growth of plants. 1
Reason (R): Its cells are actively dividing.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The species is the basic unit of classification. 1
Reason (R): Members of a species can interbreed to produce fertile offspring.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do cells divide when they reach a certain size? Use the surface area to volume ratio for a cube of side 1 cm and 3 cm. 2

17. Explain why wood is made mainly of xylem and why it is strong. 2

18. Why do scientists keep revising the classification of organisms? Give two reasons. 2

19. A cell in the human body has 46 chromosomes. State the number of chromosomes in a muscle cell, sperm, egg and zygote. 2

20. The average albedo of the Earth is 0.3. Calculate the radiation absorbed per square metre if the average incident is 342 W/m squared. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A cube of potato of mass 10 g is placed in 0.5 per cent, 1 per cent and 5 per cent sugar solutions. Predict the changes with the reasoning and mention the solution that is isotonic if the mass is unchanged in the 1 per cent solution. 3

22. A potato disc and a carrot disc are placed in iodine solution. Predict the observation and identify the tissue involved. 3

23. Construct a simple key (two choices at each step) to identify the following: pigeon, frog, shark, bat, snake. 3

24. A woman has a 30-day menstrual cycle with menstruation starting on day 1. Calculate the probable day of ovulation, and the fertile period (taking the egg life as 1 day and sperm life as 3 days). 3

OR

A household uses 250 units per month. Using 0.8 kg CO_2 per unit, calculate the annual emission and the number of trees (each absorbing 22 kg per year) needed to offset it. 3

25. A researcher wants to isolate pure mitochondria from liver cells. Describe the method in outline (centrifugation). 3

OR

State the reasons for the huge numbers of mitochondria in the cardiac muscle cells. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Anil keeps dried raisins in plain water overnight and finds them swollen next morning. He also weighs two equal potato strips of 100 g each: one kept in a 10 per cent salt solution weighs 90 g after an hour, and the other kept in distilled water weighs 112 g. 5

(i) Define osmosis.

(ii) Why do the raisins swell?

(iii) Which strip is in the hypertonic solution?

(iv) Find the percentage change in mass of each strip and explain the result.

27. Design a taxonomic key and a table for the identification of 8 organisms: rose, wheat, fern, moss, pine, yeast, bacterium and Amoeba, listing the characters used at each step. 5

OR

Describe in detail an experiment to show that the seed germination needs water, air and a suitable temperature, with the set-up of 4 test tubes, controls and conclusions. 5

END OF QUESTION PAPER — Total: 50 Marks